

ACE Critical Care

Arma Reforger Medical Expansion

Tactical Critical Care Sandbox

A vision-driven production plan for advanced prehospital emergency medicine, ALS/BLS gameplay, trauma care and critical care systems in Arma Reforger.

Version:	1.0
Status:	Concept / Pre-Production
Project Type:	Advanced Medical Gameplay Expansion for Arma Reforger
Target Platform:	Arma Reforger (Enfusion Engine)
Base Framework:	ACE Anvil Medical System

Gameplay First

Multiplayer Focused

Prehospital Critical Care

Modular Architecture



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Executive Summary

Project Summary

ACE Critical Care (ACE-CC) is a professional-grade medical gameplay expansion for Arma Reforger focused on tactical emergency medicine, prehospital critical care, airway management, trauma treatment and advanced resuscitation.

The project extends the existing ACE Anvil medical framework instead of replacing it. The core philosophy is to create believable, immersive and gameplay-relevant emergency medicine systems without turning the game into a full hospital simulator.

The project focuses on:

- Tactical emergency medicine
- Teamplay-oriented treatment chains
- Role-based medical gameplay
- Realistic but playable intervention systems
- High-quality medical devices and interactions
- Expandable modular architecture

The project intentionally avoids excessive micromanagement, unnecessary realism and ICU-level simulation complexity.

Ambition

ACE Critical Care aims to become the leading advanced medical framework for Arma Reforger.

Vision Statement

Vision Statement

“Create the most immersive and gameplay-oriented prehospital emergency medicine experience for Arma Reforger while maintaining accessibility, tactical relevance and multiplayer stability.”

Clinical & Gameplay Philosophy

Clinical Inspiration Philosophy

ACE Critical Care is inspired by modern prehospital emergency medicine, advanced life support workflows and European resuscitation principles.

The project intentionally adapts real-world emergency medicine into gameplay-oriented systems instead of reproducing medicine as a full clinical simulator.

The goal is to create believable medical decision-making, tactical pressure and meaningful interventions while remaining accessible and playable in multiplayer environments.

The Project Focuses On

- believable physiological consequences
- authentic emergency workflows
- tactical decision making
- role specialization
- teamwork under pressure
- emergency stabilization

The Project Does Not Aim To Simulate

- complete hospital medicine
- ICU-level diagnostics
- exact medication calculations
- full cardiology interpretation
- certification-grade clinical training

ALS/BLS Gameplay Philosophy

The project differentiates between BLS and ALS gameplay capabilities.

Role	Capabilities	Limitations / Focus
Soldier / Basic Infantry	CPR, AED usage, basic bleeding control, basic oxygenation, bag-valve-mask ventilation	No advanced airway procedures, no advanced medications, no manual rhythm interpretation, no invasive procedures
Medic / Combat Medic	Advanced airway management, IV/IO access, manual defibrillation, medication administration, needle decompression, ventilator usage	ALS-focused battlefield stabilization
Physician / Critical Care Specialist	Rapid sequence induction, endotracheal intubation, surgical airway, thoracic drainage, perfusor management, advanced critical care stabilization	Highest level of prehospital critical care

This role separation is intended to encourage teamwork, triage gameplay, tactical coordination, CASEVAC and MEDEVAC scenarios and realistic responsibility distribution.

Simulation Boundary Statement**ACE Critical Care Simulates**

- medical consequences
- emergency priorities
- trauma management
- airway compromise
- oxygenation and ventilation
- resuscitation workflows
- tactical emergency medicine

ACE Critical Care Does Not Simulate

- complete clinical diagnostics
- full ICU management
- exact pharmacology mathematics
- laboratory workflows
- mandatory cable management
- excessive interaction depth
- unnecessary medical micromanagement

Core Design Philosophy

Gameplay First

The player experience has priority over raw realism.

The System Should Create

- Stress
- Decision making
- Team dependency
- Prioritization
- Triage
- Role specialization
- Tactical medical gameplay

The System Should Not Create

- Excessive UI interaction
- Hospital simulator complexity
- Long repetitive procedures
- Mandatory micromanagement

Medical Design Philosophy

Realistic Logic, Simplified Execution

Medical Action Design

Medical actions should have believable consequences, follow real emergency medicine principles, reward proper treatment and punish incorrect priorities.

But actions must remain fast, understandable, multiplayer friendly, learnable and combat-compatible.

Project Objectives

ID	Objective	Description
01	Advanced Airway Management	Implement realistic airway compromise and airway intervention systems.
02	Critical Care Monitoring	Implement advanced patient monitoring systems inspired by modern prehospital monitors.
03	Trauma Expansion	Implement realistic trauma complications and advanced treatment options.
04	Advanced Resuscitation	Implement ALS/BLS workflows including manual defibrillation and advanced CPR.
05	Role-Based Medical Gameplay	Create meaningful gameplay differences between soldiers, medics and physicians.
06	Multiplayer Stability	Ensure all systems work reliably in multiplayer environments.
07	Modular Expandability	Create a clean architecture for future expansion.

Non-Goals

Medical Non-Goals

- Full ICU simulation
- Full anesthesia simulation in V0.1
- Blood gas analysis
- Laboratory systems
- Pediatric-specific medicine systems
- Exact mg/kg medication simulation
- Real-world certification-grade medicine simulation

Gameplay Non-Goals

- Cable simulation
- Mandatory ECG lead placement
- Manual syringe preparation simulation
- Excessive interaction depth
- Hyper-realistic equipment maintenance
- 8-click treatment workflows

Technical Non-Goals

- Replacing ACE core systems
- Monolithic code architecture
- Client-authoritative medical logic
- Single massive god classes
- Features without multi-player testing

Product Scope

Included Systems

System Area	Included Features
Medical Monitoring	ECG monitoring, heart rate, SpO2, NIBP, invasive blood pressure, capnography, manual defibrillation, cardioversion, pacing (later milestone)
Airway Management	Airway obstruction, manual airway clearing, suctioning, head tilt/chin lift, Guedel airway, laryngeal tube, endotracheal intubation, surgical airway
Ventilation	Bag-valve-mask ventilation, oxygen therapy, mechanical ventilation, ventilation alarms, basic ventilator modes
Trauma	Pneumothorax, tension pneumothorax, needle decompression, thoracic drainage, fracture stabilization, pain systems
Medication System	Emergency medications, side effects, drug interactions, sedation, analgesia, vasopressors
Circulation Access Resuscitation	IV access, failed IV attempts, IO access, perfusor systems ALS workflows, BLS workflows, AEDs, manual defibrillation, LUCAS device

Stakeholders

Stakeholder	Interest
Players	Immersive gameplay
MilSim Communities	Tactical realism

Medical Personnel	Authenticity
Mod Developers	API and extensibility
Mission Makers	Scenario creation
Project Team	Stable maintainable architecture

Technical Foundation

Engine

Arma Reforger – Enfusion Engine

Framework Dependencies

- ACE Anvil Medical
- Reforger Medical Components
- Enfusion Entity System
- Reforger UI Framework

Modding Areas

- Script Components
- Animation
- Prefabs
- Audio
- UI Systems
- Interaction Systems
- Replication
- Inventory Systems

ACE Anvil Integration Strategy

Architectural Philosophy

ACE Critical Care does not replace ACE Anvil Medical.

Instead, ACE Critical Care extends the ACE Anvil medical framework through modular gameplay layers, advanced physiology systems and additional emergency medicine mechanics.

ACE Anvil remains responsible for:

- baseline medical entity handling
- damage registration
- unconsciousness foundations
- bleeding state handling
- core multiplayer replication
- patient ownership logic

ACE Critical Care extends:

- airway management
- oxygenation systems
- ventilation systems
- ALS/BLS workflows
- advanced monitoring
- trauma systems
- medication systems
- advanced physiological states

Extension Philosophy

Non-Destructive Architecture

Whenever possible, ACE Critical Care extends existing ACE Anvil systems instead of replacing them entirely.

The project intentionally prioritizes:

- inheritance over overwriting
- event-driven extensions
- isolated modular systems
- compatibility-oriented architecture
- reusable gameplay components

This minimizes:

- compatibility issues
- upstream update conflicts
- multiplayer instability
- long-term maintenance complexity

Core Integration Layers

Layer	Responsibility
ACE Base Medical	Core patient state and damage foundation
ACECC Physiology	Advanced vital signs and physiology logic
ACECC Airway	Airway compromise and airway management
ACECC Ventilation	Oxygenation and ventilation systems
ACECC Monitoring	ECG, SpO2, ETCO2 and monitoring devices
ACECC Trauma	Advanced trauma complications and treatment

ACECC Medication	Drug systems and medication effects
ACECC Networking	Replication extensions and synchronization
ACECC UI	Monitor interfaces and interaction systems

Planned ACE Anvil Touchpoints

ACE System	ACE Critical Care Extension
Bleeding Systems	Hemodynamic instability and shock systems
Breathing Systems	Advanced airway and oxygenation logic
Unconsciousness	Sedation and advanced neurological states
Damage Pipeline	Organ-related trauma extensions
Medical Actions	ALS and critical care interventions
Patient State	Extended physiology and monitoring

Multiplayer Philosophy

Server Authoritative Medical Logic

All critical medical systems remain server authoritative to minimize desynchronization and gameplay exploits.

Critical synchronized systems include:

- airway states
- active monitor values
- oxygenation
- CPR states
- unconsciousness
- cardiac rhythms
- medication effects
- ventilation states

Clients primarily display information, request interactions and render local effects.

Compatibility Goals

Long-Term Compatibility Goal

ACE Critical Care aims to remain compatible with upstream ACE Anvil updates whenever technically feasible.

The project intentionally avoids deep engine-level rewrites and hardcoded overrides unless absolutely necessary.

ACE Integration Risks

Risk	Severity	Mitigation
ACE upstream breaking changes	Critical	Minimize hard overrides and isolate extension layers

Replication incompatibilities	High	Early multiplayer testing and dedicated replication architecture
Tight coupling to ACE internals	High	Maintain modular interfaces and abstraction layers
Performance overhead	Medium	Isolated update loops and scalable systems

High-Level System Architecture

```
ACECC_Core
|-- ACECC_Airway
|-- ACECC_Breathing
|-- ACECC_Monitor
|-- ACECC_Medication
|-- ACECC_Trauma
|-- ACECC_Defib
|-- ACECC_Ventilator
|-- ACECC_UI
|-- ACECC_Assets
'-- ACECC_API
```

Development Strategy

Modular Incremental Delivery

The project will use phased milestone-based delivery.

Each milestone must be playable, multiplayer tested, have isolated deliverables and avoid unfinished dependencies.

Production Pipeline

Phase	Goals	Deliverables
Pre-Production	System architecture, scope definition, core gameplay loop, technical feasibility, UI concepts, asset list, risk analysis	Game Design Document, Technical Design Document, Medical Logic Overview, Asset Planning Sheet, Milestone Plan
Prototype Phase	Validate core gameplay loops, test ACE integration, validate replication, validate interaction systems	Prototype airway system, prototype monitor UI, prototype medication system, prototype trauma workflow
Vertical Slice	Deliver one fully functional gameplay scenario	Fully integrated medical workflow, multiplayer validation, usability validation
Production Phase	Feature expansion, asset production, system stabilization, documentation	Playable milestone builds

Release & Launch	Post-Community testing, bug fixing, balancing, API support, feature roadmap	Public release and post-launch roadmap
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Vertical Slice Example

Polytrauma patient with airway compromise, tension pneumothorax, shock, analgesia, intubation, decompression and transport ventilation.

Milestone Roadmap

Milestone M0 – Foundation & Preproduction

Duration: 4–6 Weeks

Objectives

- Establish project architecture
- Create planning documentation
- Define medical gameplay scope
- Establish coding conventions
- Setup repository and workflows

Work Package	Tasks	Deliverables
WP-M0-01 Project Infrastructure	Repository setup, branching strategy, build pipeline, mod structure, naming conventions, folder architecture	Git repository, build structure, base mod package
WP-M0-02 Technical Architecture	Component architecture, replication strategy, state machine planning, medical event pipeline	Technical Design Document, architecture diagrams
WP-M0-03 Medical System Design	Airway logic design, breathing logic design, trauma logic design, medication interaction design	Medical Logic Specification
WP-M0-04 Asset Planning	Device inventory, 3D asset categorization, external asset review, poly budget definition, texture standards	Asset backlog, production standards
WP-M0-05 ACE Integration Architecture	Dependency structure, extension strategy, override policy, modular integration planning	ACE integration specification
WP-M0-06 Multiplayer Authority Design	Ownership logic, replication authority, synchronization planning	Multiplayer authority architecture
WP-M0-07 Event & Hook Mapping	ACE hook analysis, event integration planning, action extension mapping	ACE event integration document

Milestone M1 – Core Medical Framework

Duration: 6–8 Weeks

Objectives

- Implement core state systems
- Establish patient status framework
- Implement baseline UI systems

Work Package	Tasks	Deliverables
WP-M1-01 Core State Machine	Airway states, breathing states, consciousness states, monitoring hooks	Core medical state system
WP-M1-02 Replication Layer	Multiplayer synchronization, event replication, prediction handling, state ownership	Multiplayer-safe replication layer
WP-M1-03 Base UI Framework	Monitor UI architecture, device interaction UI, alert systems, notification systems	Base UI framework
WP-M1-04 Basic Audio Framework	Monitor beeps, alarm system, defibrillator sounds, airway sounds	Audio interaction framework

Milestone M2 – Airway & Ventilation

Duration: 8–10 Weeks

Objectives

- Deliver advanced airway gameplay
- Deliver oxygenation mechanics
- Deliver ventilation systems

Work Package	Tasks	Deliverables
WP-M2-01 Airway Obstruction System	Blood obstruction, vomit obstruction, tongue obstruction, airway compromise logic	Dynamic airway obstruction system
WP-M2-02 Manual Airway Management	Head tilt mechanics, manual airway clearing, suction system	Interactive airway clearing system
WP-M2-03 Airway Devices	Guedel airway, laryngeal tube, endotracheal tube, surgical airway	Functional airway devices
WP-M2-04 Ventilation System	BVM ventilation, oxygen masks, ventilator logic, ventilation parameters	Ventilation framework
WP-M2-05 Capnography	ETCO2 logic, waveform rendering, tubus verification	Functional capnography system

Milestone M3 – Monitoring & Defibrillation

Duration: 6–8 Weeks

Objectives

- Deliver advanced patient monitoring
- Deliver ALS defibrillation workflows

Work Package	Tasks	Deliverables
WP-M3-01 Monitor Device	Device prefab, screen UI, interaction logic, carry system	Functional monitor device
WP-M3-02 ECG System	ECG display, rhythm states, waveform generation	ECG visualization system
WP-M3-03 Defibrillation	Manual defib, shock charging, rhythm analysis, shock safety checks	Advanced defibrillation system
WP-M3-04 AED Workflow	Automated rhythm detection, guided CPR workflow	Soldier-level AED system

Milestone M4 – Trauma & Critical Care

Duration: 8–10 Weeks

Objectives

- Deliver advanced trauma gameplay
- Deliver critical care procedures

Work Package	Tasks	Deliverables
WP-M4-01 Pneumothorax System	Pneumothorax states, pressure buildup, respiratory compromise	Thoracic injury system
WP-M4-02 Needle Decompression	Decompression logic, temporary relief mechanics	Needle decompression gameplay
WP-M4-03 Thoracic Drainage	Drain placement, drain management, lung recovery mechanics	Thoracic drainage system
WP-M4-04 Fracture System	Fracture pain, splinting, mobility effects	Fracture stabilization system

Milestone M5 – Medication & Circulation

Duration: 8 Weeks

Objectives

- Deliver medication gameplay
- Deliver IV/IO systems

Work Package	Tasks	Deliverables
WP-M5-01 Medication Framework	Drug effects, side effects, timed pharmacology, interaction logic	Medication effect system

WP-M5-02 IV Access	IV placement, failed attempts, access quality	IV access system
WP-M5-03 IO Access	IO placement, emergency access logic	IO system
WP-M5-04 Perfusor System	Continuous medication delivery, rate configuration, medication consumption	Perfusor gameplay system

Milestone M6 – Polish & Release

Duration: 6 Weeks

Objectives

- Stabilization
- Optimization
- QA
- Public release

Work Packages

- WP-M6-01 Performance Optimization
- WP-M6-02 Multiplayer Stress Testing
- WP-M6-03 UI Polish
- WP-M6-04 Documentation
- WP-M6-05 Steam Workshop Release

3D Asset Production Plan

Asset Categories

Category	Assets
Medical Devices	Monitor, defibrillator, ventilator, perfusor, suction device, AED, LUCAS device
Airway Equipment	Guedel airway, laryngeal tube, endotracheal tube, laryngoscope, cricothyrotomy kit
Trauma Equipment	Chest decompression kit, thoracic drainage kit, splints, oxygen systems

3D Production Pipeline

Phase	Goals	Deliverables
Phase 1 – Blockout	Scale validation, ergonomic validation, animation validation	Greybox models
Phase 2 – High Poly	Detail modeling, mechanical fidelity	High poly assets

Phase 3 – Game Ready Assets	Optimization, UVs, baking, texturing	Production-ready assets
Phase 4 – Engine Integration	Prefabs, collision, interaction points, animation hooks	Functional in-game assets

3D Asset Standards

Asset Type	Triangle Budget
Small prop	2k–8k
Medium device	10k–25k
Hero device	30k–60k

Texturing Standards

Type	Standard
Workflow	PBR
Texture Size	2K standard
Hero Assets	4K optional
LODs	Mandatory

Scripting Standards

Coding Philosophy

DO

- Use modular systems
- Use event-driven architecture
- Separate gameplay and UI logic
- Keep systems independent
- Document systems
- Use reusable components
- Build multiplayer-first

DO NOT

- Hardcode gameplay logic into UI
- Create god classes
- Use duplicated logic
- Ignore replication
- Ignore performance
- Build singleplayer-only systems

Coding Conventions

Naming

```
ACECC_ComponentName  
ACECC_MonitorComponent  
ACECC_AirwayStateComponent
```

Separation

Gameplay Logic
UI Logic
Replication Logic
Animation Logic
Audio Logic

Must remain isolated.

Multiplayer Strategy

Core Rule

The server is authoritative.

Clients only display information, request actions and play local effects.

The server validates actions, controls medical state and synchronizes results.

Risk Analysis

Risk	Severity	Mitigation
Scope explosion	Critical	Strict milestone control
Multiplayer desync	Critical	Early replication testing
Overcomplex gameplay	High	Gameplay-first reviews
UI overload	High	Minimalist interaction design
Performance problems	High	Modular optimization
3D asset bottlenecks	Medium	Reuse + staged production
Medical realism conflicts	Medium	Simplified logic philosophy

Resource Planning

Role Area	Responsibilities
Programming	Gameplay systems, UI systems, networking, tooling
Art	Hard surface modeling, texturing, animation, UI/UX
Design	Medical gameplay design, balancing, testing
QA	Multiplayer testing, medical workflow testing, regression testing

Estimated Development Complexity

Area	Complexity
Airway systems	High
Monitoring UI	High
Defibrillation	Medium

Ventilator	High
Trauma systems	Medium
Medication logic	Medium
Multiplayer synchronization	Critical
3D asset pipeline	High

Medical Sandbox Vision

Tactical Critical Care Sandbox

ACE Critical Care is designed as a tactical critical care sandbox. The system is intended to create emergent medical gameplay scenarios rather than scripted treatment chains.

Examples include:

- prolonged field care
- tactical casualty extraction
- multi-casualty incidents
- CASEVAC and MEDEVAC scenarios
- prolonged transport stabilization
- resource-limited battlefield medicine

The goal is to create a dynamic environment where player decisions, teamwork and prioritization directly influence patient outcomes.

Recommended MVP Scope

Asset Production Strategy

Because high-quality medical devices are central to immersion and gameplay readability, the project prioritizes hero assets over excessive prop quantity.

Initial Priority Assets

- Patient monitor
- Defibrillator
- Ventilator
- Suction unit
- AED
- Airway devices

Asset Scope Control Rules

DO

- Reuse existing assets where appropriate
- Prioritize gameplay-critical devices
- Focus on recognizable silhouettes
- Use staged asset production
- Create scalable prefab systems

DO NOT

- Overproduce small decorative props early
- Spend excessive time on hidden details
- Build ultra-high-poly assets without gameplay value
- Delay gameplay implementation for visual

perfection

MVP Features

Included

- Basic monitor
- SpO2
- HR
- Capnography
- Airway obstruction
- Suction
- Guedel airway
- Laryngeal tube
- ETI
- BVM ventilation
- Pneumothorax
- Needle decompression
- Manual defibrillation

Deliberately Excluded From Initial MVP

- Advanced ventilator modes
- Thoracic drainage
- Perfusor systems
- Complex medication interactions
- Full ECG interpretation
- Advanced pacing
- Full sedation workflows
- Extended ICU systems

Excluded From MVP

- Advanced ventilator modes
- Full pacing system
- Complex perfusors
- Pediatric systems
- Full anesthesia
- Advanced ECG interpretation

Long-Term Expansion Possibilities

- Medical evacuation systems
- ICU transport gameplay
- Blood products
- Advanced ventilator modes
- Mission maker APIs
- AI medic support
- Dynamic triage systems
- Medical training scenarios
- Campaign integration

Quality Standards

Quality Area	Standards
Technical Quality	Stable multiplayer behavior, low desync rates, consistent FPS impact, modular architecture
Gameplay Quality	Fast interactions, intuitive workflows, meaningful decisions, high immersion
Visual Quality	Consistent medical art direction, authentic device appearance, professional UI design

Final Strategic Recommendation

Strategic Recommendation

The success of ACE Critical Care will not depend on creating the most realistic medical simulation. It will depend on prioritizing gameplay, maintaining clarity, delivering polished systems, controlling project scope, releasing stable milestones and building strong multiplayer experiences.

The project should evolve incrementally.

A smaller polished release is significantly more valuable than a massive unfinished system.

Final Positioning

ACE Critical Care has the potential to become the defining advanced medical gameplay framework for Arma Reforger if development remains focused, modular and gameplay-driven.